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Algebraic Number Theory 3

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1. Reciprocity laws

The main goal of this course is to generalize the Legendre symbol and its reciprocity laws. change the motivation for the course, this one is quite bad.

Definition 1.0.1 (Legendre symbol)

Let p be a prime and $a \in \mathbb{Z}$ an integer coprime with p. The *Legendre symbol* is defined as

$$\left(\frac{a}{p}\right) := \begin{cases} 1 & \text{if } x^2 \equiv a \bmod p \text{ is solvable} \\ -1 & \text{otherwise} \end{cases}$$

Properties

- *(Euler)* The Legendre symbol is multiplicative in a and for $p > 2$ it holds
$$\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}} \bmod p.$$
- *(Gauss)* It satisfies the following reciprocity law: if $a \neq b$ are odd primes then
$$\left(\frac{a}{b}\right)\left(\frac{b}{a}\right) = (-1)^{\frac{a-1}{2}\frac{b-1}{2}}.$$
- In particular we have
$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}, \qquad \left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}.$$

By generalizing the symbol and its properties to any power studying the absolute Galois group turns out to be foundamental.

1.1. Hilbert's reinterpretation

Hilbert generalized the Legendre symbol, originally defined for integers and more generally for rational numbers, to the completions of $\mathbb{Q}$: the p-adic fields $\mathbb{Q}_p$ and the real field $\mathbb{Q}_\infty = \mathbb{R}$.

Definition 1.1.2 (Hilbert symbol)

For any $p \in \mathbb{P} \cup \{\infty\}$ let $a, b \in \mathbb{Q}_p$. The *Hilbert symbol* is defined as

$$(a,b)_p := \begin{cases} 1 & \text{if } ax^2 + by^2 = 1 \text{ is solvable in } \mathbb{Q}_p \\ -1 & \text{otherwise} \end{cases}$$

Lemma 1.1.1

The following are equivalent:
1. The Hilbert symbol has value $(a,b)_p = 1$;
2. The equation $z^2 - ax^2 - by^2 = 0$ has a non-trivial solution;
3. The element $a \in \mathbb{Q}_p$ is the norm of an element in $\mathbb{Q}_p\left(\sqrt{b}\right)$;
4. The element $b \in \mathbb{Q}_p$ is the norm of an element in $\mathbb{Q}_p(\sqrt{a})$.

Here the norm $N : \mathbb{Q}_p(\sqrt{a}) \to \mathbb{Q}_p$ is the map

$$\begin{cases} \mathrm{id}_{\mathbb{Q}_p} & \text{if } a \in \mathbb{Q}_p^2 \\ x + y\sqrt{a} \mapsto x - y\sqrt{a} & \text{if } a \notin \mathbb{Q}_p^2 \end{cases}$$

In general if $L|K$ is a finite Galois extension with group G, the norm is defined as $N(x) = \sum_{\sigma \in G} \sigma x$ for each $x \in L$.

Proof: We only prove $(2) \Leftrightarrow (3)$ since the rest is trivial. When $b \in \mathbb{Q}_p^2$ both conditions are always true, from now on suppose $b \notin \mathbb{Q}_p^2$. If (x, y, z) is a solution to the equation, so is $(1, y/x, z/x)$ and we get $a = (z/x)^2 - b(y/x)^2 = N\left(z/x + \sqrt{b} \cdot y/x\right)$. Conversely, if $a = N\left(z + \sqrt{b} \cdot y\right)$ for some $y, z \in \mathbb{Q}_p$ then the touple $(1, y, z)$ is a solution to the equation. □

Proposition 1.1.1

If $p \neq 2, \infty$ then for any $a, b \in \mathbb{Z}$ we have

$$\begin{aligned}(a,b)_p &= (-1)^{\frac{p-1}{2} v_p(a) v_p(b)} \cdot \left(\frac{a}{p}\right)^{v_p(b)} \cdot \left(\frac{b}{p}\right)^{-v_p(a)} \\ &= \left[\overline{(-1)^{v_p(a)v_p(b)} \cdot a^{v_p(b)} \cdot b^{-v_p(a)}}\right]^{\frac{p-1}{2}}\end{aligned}$$

where $v_p(a) = \max\{n \in \mathbb{N} : p^n \mid a\}$ is the p-adic valuation of a and $a \mapsto \overline{a}$ is reduction modulo p.

In particular if $p \nmid a$ it simplifies to $(a, b)_p = \left(\frac{a}{p}\right)^{v_p(b)}$.

Lemma 1.1.2

Let $a, b, c \in \mathbb{Q}_p^\times$. We have:
1. $(1, c)_p = 1$;
2. $(-c, c)_p = 1$;
3. $(a, c)_p = 1 \Longrightarrow (ab, c)_p = (a, c)_p (b, c)_p$;
4. $(c, c)_p = (-1, c)_p$.

Proof: Let $N : \mathbb{Q}_p(\sqrt{c}) \to \mathbb{Q}_p$ be the norm. We are going to repeatedly apply Lemma 1.1.1.
1. We have $1 = N(1)$.
2. If $c \in \mathbb{Q}_p^{\times 2}$ is a square, then $N = \mathrm{id}_{\mathbb{Q}_p}$, so $-c = N(-c)$. Otherwise $-c = N(c)$.
3. If a and b are norms we have

   $$a = N(x), b = N(y) \Longrightarrow ab = N(xy), a^{-1} = N(x^{-1}).$$

   Therefore if $(a, c)_p = 1$ (i.e. $a = N(x)$) then $b = N(y) \Leftrightarrow ab = N(xy)$.
4. Notice that by definition of Hilbert's symbol it is invariant under multiplication of either argument by a square. Hence

   $$(c, c)_p = (-c, c)_p (c, c)_p = \left(-c^2, c\right)_p = (-1, c)_p.$$

□

Lemma 1.1.3

If $p > 2$ and $u, v \in \mathbb{Z}_p^\times$ then $(u, v)_p = 1$.

Proof: Let $\overline{u} = u \bmod p$ and $\overline{v} = v \bmod p$, we claim $1 - \overline{u}x^2 - \overline{v}y^2 = 0$ has a trivial solution in $\mathbb{F}_p$. Indeed, since $\mathbb{F}_p^\times$ is cyclic of order $p-1$ it contains $\frac{p-1}{2}$ squares. Adding back 0 we get

$$\left|\left\{x^2 : x \in \mathbb{F}_p\right\}\right| = \frac{p+1}{2} \implies \left|\left\{\overline{u}x^2 : x \in \mathbb{F}_p\right\}\right| = \left|\left\{1 - \overline{v}y^2 : y \in \mathbb{F}_p\right\}\right| = \frac{p+1}{2}.$$

By a counting argument these two sets have an element in common, yielding the desired solution.

We conclude by applying Proposition 1.1.2 to lift the roots we found to $\mathbb{Z}_p$. □

Proposition 1.1.2 (Hensel's lemma)

If we take a polynomial $f \in \mathbb{Z}_p[x]$ and a p-adic number $a_0 \in \mathbb{Z}_p$ such that

$$f(a_0) \equiv 0 \bmod p, \qquad f'(a_0) \not\equiv 0 \bmod p$$

then there exist a p-adic number $a \in \mathbb{Z}_p$ such that $f(a) = 0$ and $a \equiv a_0 \bmod p$. In other words, under these conditions roots can be lifted from $\mathbb{F}_p$ to $\mathbb{Z}_p$.

Proposition 1.1.3

Let $p > 2$ and $a, b \in \mathbb{Q}_p^\times$. Write them as $a = p^\alpha u, b = p^\beta v$ with $u, v \in \mathbb{Z}_p$, we have

$$(a,b)_p = (-1)^{\alpha\beta\frac{p-1}{2}} \left(\frac{\overline{u}}{p}\right)^{-\beta} \left(\frac{\overline{v}}{p}\right)^{\alpha}.$$

Proof: Since $(a,b)_p$ only depends on the classes $a, b \bmod \mathbb{Q}_p^{\times 2}$ we may assume $\alpha, \beta \in \{0,1\}$. By symmetry we only need to check 3 cases:

- $\boxed{\alpha = \beta = 0}$ Apply Lemma 1.1.3.
- $\boxed{\alpha = 1, \beta = 0}$ We need to show $(pu, v)_p = \left(\frac{\overline{v}}{p}\right)$. We have

  $$(pu,v)_p \overset{\text{Lemma 1.1.3}}{=} (pu,v)_p \cdot (u^{-1},v)_p = (p,v)_p.$$

  If $v \in \mathbb{Q}_p^{\times 2}$ is a square, then $(p,v)_p = (p,1)_p = 1 = \left(\frac{\overline{v}}{p}\right)$. Otherwise, the equation $z^2 - px^2 - y^2 = 0$ has no non-trivial solutions in $\mathbb{F}_p$. By homogeneity it has no non-trivial solutions in $\mathbb{Q}_p$ either, that is $(p,v)_p = -1$. By Proposition 1.1.2 we also have $\left(\frac{\overline{v}}{p}\right) = -1$ (if $\overline{v}$ was a square in $\mathbb{F}_p^\times$ it could be lifted to a square in $\mathbb{Z}_p^\times$).
- $\boxed{\alpha = \beta = 1}$ We need to show $(pu, pv)_p = (-1)^{\frac{p-1}{2}} \left(\frac{\overline{u}}{p}\right)\left(\frac{\overline{v}}{p}\right)$. We have

  $$(pu,pv)_p \overset{\text{Lemma 1.1.2}}{=} (pu,pv)_p(-pv,pv)_p = (-uv,pv)_p.$$

  Since $p \nmid uv$ we can apply the previous case and get

  $$(-uv,pv)_p = \left(\frac{-\overline{uv}}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{\overline{u}}{p}\right)\left(\frac{\overline{v}}{p}\right) = (-1)^{\frac{p-1}{2}}\left(\frac{\overline{u}}{p}\right)\left(\frac{\overline{v}}{p}\right).$$

□

Corollary 1.1.1

When $p > 2$ we have $(a,c)_p(b,c)_p = (ab,c)_p$ for every $a, b, c \in \mathbb{Q}_p^\times$.

Proof: Let $a = p^\alpha u, b = p^\beta v, c = p^\gamma w$ with $u, v, w \in \mathbb{Z}_p^\times$. We have

$$\begin{aligned}(a,c)_p(b,c)_p &= (-1)^{\alpha\gamma\frac{p-1}{2}}\left(\frac{u}{p}\right)^{-\gamma}\left(\frac{w}{p}\right)^{\alpha}\cdot(-1)^{\beta\gamma\frac{p-1}{2}}\left(\frac{v}{p}\right)^{-\gamma}\left(\frac{w}{p}\right)^{\beta}\\ &= (-1)^{(\alpha+\beta)\gamma\frac{p-1}{2}}\left(\frac{uv}{p}\right)^{-\gamma}\left(\frac{w}{p}\right)^{\alpha+\beta}\\ &= (ab,c)_p\end{aligned}$$

□

Proposition 1.1.4

Let $a,b\in\mathbb{Q}_2^\times$ and write them as $a=2^\alpha u, b=2^\beta v$ with $u,v\in\mathbb{Z}_2^\times$. Define

$$\varepsilon(u) := \frac{u-1}{2} \bmod 2, \qquad \omega(u) := \frac{u^2-1}{8} \bmod 2.$$

The 2-adic Hilbert symbol has value

$$(a,b)_2 = (-1)^{\varepsilon(u)\varepsilon(v)+\alpha\omega(v)+\beta\omega(u)}.$$

It is a bimultiplicative function.

Theorem 1.1.1 (Hilbert's reciprocity law)

For any $a,b\in\mathbb{Q}^\times$ we have

$$\prod_p (a,b)_p = 1.$$

Proof: Hilbert's symbol is trivial for all but finitely many p, hence the product is well defined. Using multiplicativity in both entries we reduce to the case where both a and b are either -1 or prime.

- $\boxed{a=b=-1}$ *Exercise.*
- $\boxed{a=b=q}$ We reduce to $(q,q)_p=(-1,q)_p$. *Exercise.*
- $\boxed{a=2, b>2}$ If $p\notin\{2,b\}$ then $(2,b)_p=1$ by Proposition 1.1.3. Otherwise,

$$(2,b)_2 = (-1)^{\omega(b)} = \left(\frac{2}{b}\right) = (2,b)_b.$$

- $\boxed{a>b>2}$ If $p\notin\{2,a,b\}$ then $(a,b)_p=1$. Othewise,

$$(a,b)_2 = (-1)^{\frac{a-1}{2}\frac{b-1}{2}}, \qquad (a,b)_b = \left(\frac{a}{b}\right), \qquad (a,b)_a = \left(\frac{b}{a}\right),$$

  and by Gauss's reciprocity law we conclude.

□

Hilbert's reciprocity law incorporates Gauss's quadratic reciprocity. Indeed, if $a,b>0$:

- $p\nmid 2ab \implies (a,b)_p=1$;
- $2\nmid ab \implies (a,b)_2=(-1)^{\frac{a-1}{2}\frac{b-1}{2}}$;
- $(aa',b)=(a,b)(a',b)$.

Taking a,b as distinct primes recovers Gauss's law. Hilbert's and Gauss's reciprocity laws are in fact equivalent statements.

1.2. Hilbert's 9th problem

Next, Hilbert proceeded to generalize the result to n-powers in arbitrary number fields.

Let's consider at first the finite extensions $K|\mathbb{Q}$ contatining the n-th roots of unity $\mu_n \subset K$. Denote by $\mathcal{O}_K$ the ring of integers of K. If $\mathfrak{p} \subset \mathcal{O}_K$ is a prime ideal the goal is to define a generalized Hilbert symbol $(a,b)_\mathfrak{p} \in \mu_n$ such that Hilbert's reciprocity law holds, that is $\prod_\mathfrak{p} (a,b)_\mathfrak{p} = 1$ for every $a, b \in K^\times$.

Definition 1.2.3 (Generalized Hilbert symbol in the presence of roots of unity)

We set

$$(a,b)_\mathfrak{p} := \left[\overline{(-1)^{v_\mathfrak{p}(a)v_\mathfrak{p}(b)} \cdot a^{v_\mathfrak{p}(b)} \cdot b^{-v_\mathfrak{p}(a)}}\right]^{\frac{q-1}{n}}$$

where $\cdot \mapsto \bar{\cdot}$ is reduction $\bmod \mathfrak{p}$ and $q = |\mathcal{O}_K/\mathfrak{p}|$.

In particular if $\mathfrak{p} \nmid a$ it simplifies to $(a,b)_\mathfrak{p} = \overline{a^{\frac{q-1}{n}}}^{v_\mathfrak{p}(b)}$. If $v_\mathfrak{p}(b) = 1$, the value $(a,b)_\mathfrak{p}$ does not depend on the choice of b, and it corresponds to the unique root of unity congruent to $a^{\frac{q-1}{n}} \bmod \mathfrak{p}$.

Definition 1.2.4 (Generalized Legendre symbol)

If $\mathfrak{p} \nmid a$ pick any $b \in K^\times$ such that $v_\mathfrak{p}(b) = 1$, we set

$$\left(\frac{a}{\mathfrak{p}}\right) := (a,b)_\mathfrak{p}.$$

This generalizes the classical Legendre symbol, in fact

$$\left(\frac{a}{\mathfrak{p}}\right) \neq 1 \iff x^n = a \bmod \mathfrak{p} \text{ is solvable.}$$

Properties

- $(a,b)_\mathfrak{p} = 1 \iff b$ is a norm from $K_\mathfrak{p}(\sqrt[n]{a})$;
- $(a,b)_\mathfrak{p} = 1 \iff a$ is a norm from $K_\mathfrak{p}\left(\sqrt[n]{b}\right)$;
- $\overline{(a,b)_\mathfrak{p}}$ is multiplicative in both entries and anti-commutative.

When the roots of unity are not contained in the field $\mu_n \not\subset K_\mathfrak{p}$ the Hilbert symbol can still be defined but it won't depend on two elements $a, b \in K^\times$. Suppose for a moment $\mu_n \subset K_\mathfrak{p}, a \notin K_\mathfrak{p}^{\times n}$. The extension $K_\mathfrak{p}(\sqrt[n]{a})|K_\mathfrak{p}$ is Galois with cylic group, let $G_a := \mathrm{Gal}\big(K_\mathfrak{p}(\sqrt[n]{a})|K_\mathfrak{p}\big) \simeq \mathbb{Z}/n\mathbb{Z}$. There is an isomorphism

$$\mathrm{Hom}(G_a, \mu_n) \xrightarrow{\sim} \mathbb{Z}/n\mathbb{Z}, \qquad \sigma \mapsto \frac{\sigma(\sqrt[n]{a})}{\sqrt[n]{a}} \in \mu_n$$

and therefore we can interpret a as the generator of this group of characters.

In general we define the Hilbert symbol

$$\mathrm{Hom}\Big(\mathrm{Gal}\big(\overline{K}_\mathfrak{p}|K_\mathfrak{p}\big), \mathbb{Z}/n\mathbb{Z}\Big) \times K_\mathfrak{p}^\times \to \mathbb{Z}/n\mathbb{Z}, \qquad [\chi, b] \mapsto (\chi, b)_\mathfrak{p}.$$

In the case where $\mu_n \subset K_\mathfrak{p}$ the two symbols are effectively the same: let $\chi_a : \sigma \mapsto \frac{\sigma(\sqrt[n]{a})}{\sqrt[n]{a}}$, we have $(\chi_a, b)_\mathfrak{p} = (a, b)_\mathfrak{p}$.

Consider a character $\chi \in \mathrm{Hom}\big(\mathrm{Gal}\big(\overline{K}|K\big), \mathbb{Z}/n\mathbb{Z}\big)$, we shall see it induces a map $\chi_\mathfrak{p}$ via

$$\begin{array}{ccc} \mathrm{Gal}\big(\overline{K}_\mathfrak{p}|K_\mathfrak{p}\big)^{\mathrm{ab}} & \hookrightarrow & \mathrm{Gal}\big(\overline{K}|K\big)^{\mathrm{ab}} \\ & \searrow^{\chi_\mathfrak{p}} & \downarrow{\scriptstyle \chi} \\ & & \mathbb{Z}/n \end{array}$$

Theorem 1.2.2 (Artin's reciprocity law)

Fix an element $b \in K^\times$, we have

$$\sum_\mathfrak{p} (\chi_\mathfrak{p}, b) = 0.$$

The generalized Hilbert symbol defines a reciprocity law $K_\mathfrak{p}^\times \to \mathrm{Gal}\big(\overline{K}_\mathfrak{p}|K_\mathfrak{p}\big)^{\mathrm{ab}}$ which is "almost" an isomorphism. In particular, for every n we have an isomorphism of finite groups $K_\mathfrak{p}^\times/n \xrightarrow{\sim} \mathrm{Gal}\big(\overline{K}_\mathfrak{p}|K_\mathfrak{p}\big)^{\mathrm{ab}}/n$. To define such a map we shall contruct its dual

$$\begin{aligned} \mathrm{Hom}\Big(\mathrm{Gal}\big(\overline{K}_\mathfrak{p}|K_\mathfrak{p}\big)^{\mathrm{ab}}/n, \mathbb{Z}/n\Big) &\longrightarrow \mathrm{Hom}\big(K_\mathfrak{p}^\times/n, \mathbb{Z}/n\big), \\ \chi &\longmapsto \big[b \mapsto (\chi, b)_\mathfrak{p}\big]. \end{aligned}$$

2. Cohomology of groups

2.1. Construction of the cohomology

Definition 2.1.5 (G-modules)

Let G be a group. A *G-module* is an abelian group A equipped with a left action by G denoted by $(\sigma, a) \mapsto \sigma a$. Equivalently, this is a left module over $\mathbb{Z}[G]$.

We say a G-module A is *trivial* if $\sigma a = a$ for every $a \in A$. A *morphism of G-modules* is a homomorphism of abelian groups compatible with the G-action. These form a group denoted $\mathrm{Hom}_G(\cdot, \cdot)$.

We denote by A^G the submodule of G-invariant elements of A, that is

$$A^G := \{a \in A : \sigma a = a \text{ for every } \sigma \in G\}.$$

For every G-module A we shall define abelian groups $H^i(G, A)$, indexed by $i > 0$, satisfying:
- $H^0(G, A) = A^G$;
- Every G-morphism $A \to B$ induces a homomorphism $H^i(G, A) \to H^i(G, B)$, for every i;
- Given a short exact sequence of G-modules $0 \to A \to B \to C \to 0$ there is a long exact sequence

$$0 \to H^0(G, A) \to H^0(G, B) \to H^0(G, C) \to H^1(G, A) \to ...$$

Definition 2.1.6 (Group cohomology)

Given a G-module A, we define the i-th cohomology group of G with coefficients in A as

$$H^i(G, A) := \mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, A).$$

We now recall how the Ext functor is constructed, aiming at computing some useful group cohomologies.

Definition 2.1.7 (Complexes of modules)

A *(cochain) complex of modules*, denoted $A^\bullet$, is a sequence of modules

$$... \longrightarrow A^i \xrightarrow{d^i} A^{i+1} \xrightarrow{d^{i+1}} A^{i+2} \longrightarrow ...$$

such that every composition of maps vanishes, i.e. $d^{i+1} \circ d^i = 0$ for every i.

A *morphism of complexes* $\varphi : A^\bullet \to B^\bullet$ is a collection of homomorphisms $\varphi^i : A^i \to B^i$ such that all diagrams commute.

A *short exact sequence of complexes* is a sequence of morphisms $0 \to A^\bullet \to B^\bullet \to C^\bullet \to 0$ such that every short sequence $0 \to A^i \to B^i \to C^i \to 0$ is exact.

Definition 2.1.8 (Cohomology)

Let $A^\bullet$ be a complex of modules with boundary maps d^i. We define the group of i*-cocycles* as $\mathcal{Z}^i(A^\bullet) := \ker d^i$, the group of i*-coboundaries* as $\mathcal{B}^i(A^\bullet) := \operatorname{im} d^{i+1}$ and the group of i*-cochains* (also called the i*-th cohomology group*) as the quotient $H^i(A^\bullet) := \mathcal{Z}^i(A^\bullet)/\mathcal{B}^i(A^\bullet)$.

Proposition 2.1.5 (Long exact sequence in cohomology)

Let $0 \to A^\bullet \to B^\bullet \to C^\bullet \to 0$ be a short exact sequence of complexes. There is a long exact sequence

$$\ldots \to H^i(A^\bullet) \to H^i(B^\bullet) \to H^i(C^\bullet) \to H^{i+1}(A^\bullet) \to \ldots$$

Definition 2.1.9 (Projective modules)

A (left) R-module is *projective* if every surjective map $\alpha : A \twoheadrightarrow B$ induces a surjective map $\alpha : \operatorname{Hom}(P, A) \twoheadrightarrow \operatorname{Hom}(P, B)$. In other words, if the functor $\operatorname{Hom}(P, \cdot)$ is right exact.

The ring R as a module over itself is projective, and the direct sum of projective modules is still projective. In particular free R-modules are always projective.

Definition 2.1.10 (Projective resolutions)

Let M be a R-module. A *projective resolution of* M is an exact sequence

$$\ldots \longrightarrow P_2 \longrightarrow P_1 \longrightarrow P_0 \longrightarrow M \longrightarrow 0$$

where each P_i is projective.

To write a projective resolution as a *cochain* complex we can use the indexing $P^i = P_{-i}$.

Fact

Projective resolutions exist for any R-module, and moreover we can always find *free* resolutions for any R-module.

Let A be a G-module, and let $P_\bullet \to \mathbb{Z}$ be any projective resolution of $\mathbb{Z}$ as a trivial G-module. Applying the functor $\operatorname{Hom}_G(\cdot, A)$ yields a cochain complex $\operatorname{Hom}_G(P_\bullet, A)$, whose i-th cohomology group is by definition $\operatorname{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, A)$. Thus,

$$H^i(G, A) = \operatorname{Ext}_{\mathbb{Z}[G]}(\mathbb{Z}, A) = H^i(\operatorname{Hom}_G(P_\bullet, A)).$$

In particular we have $H^0(\mathbb{Z}, A) = \operatorname{Hom}_G(\mathbb{Z}, A) = A^G$. Indeed, any G-morphism $\varphi : \mathbb{Z} \to A$ is uniquely determined by $\varphi(1)$. Since $\mathbb{Z}$ is trivial, we must have $\varphi(1) = \varphi(\sigma 1) = \sigma\varphi(1)$ for every $\sigma \in G$, that is $\varphi(1) \in A^G$. Conversely, any choice of an element in A^G defines such a morphism.

For group cohomology to be well-defined, and more generally for the Ext functor to be so, we need to check that the cohomology groups we get do not depend on the choice of a projective resolution for $\mathbb{Z}$.

Lemma 2.1.4

Let $P_\bullet \to A$ be a projective resolution of A and $B_\bullet \to B \to 0$ an exact sequence. A morphism $\alpha : A \to B$ can be lifted to a morphism of complexes via the diagram

$$\begin{array}{ccccccccccc} \dots & \longrightarrow & P_2 & \longrightarrow & P_1 & \longrightarrow & P_0 & \longrightarrow & A & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha_2} & & \downarrow{\scriptstyle \alpha_1} & & \downarrow{\scriptstyle \alpha_0} & & \downarrow{\scriptstyle \alpha} & & \\ \dots & \longrightarrow & B_2 & \longrightarrow & B_1 & \longrightarrow & B_0 & \longrightarrow & B & \longrightarrow & 0 \end{array}$$

Moreover, the maps induced in cohomology $H^i(\alpha_i) : H^i(P_\bullet) \to H^i(B_\bullet)$ are unique.

Proof: Let $\alpha_{-1} = \alpha$ and $\alpha_{-2} = 0$, we are going to construct α_i by induction on i. Consider the diagram:

$$\begin{array}{ccccccccc} \dots & \longrightarrow & P_{i+1} & \xrightarrow{p_{i+1}} & P_i & \xrightarrow{p_i} & P_{i-1} & \longrightarrow & \dots \\ & & \downarrow{\scriptstyle \alpha_{i+1}} & \searrow{\scriptstyle \tilde{\alpha}_{i+1}} & \downarrow{\scriptstyle \alpha_i} & & \downarrow{\scriptstyle \alpha_{i-1}} & & \\ \dots & \longrightarrow & B_{i+1} & \xrightarrow[b_{i+1}]{} & B_i & \xrightarrow[b_i]{} & B_{i-1} & \longrightarrow & \dots \end{array}$$

Let $\tilde{\alpha}_{i+1} : P_{i+1} \to \operatorname{im} \alpha_i \subseteq B_i$ be the map $\tilde{\alpha}_{i+1} = \alpha_i \circ p_{i+1}$. Since

$$b_i \circ \tilde{\alpha}_{i+1} = b_i \circ \alpha_i \circ p_{i+1} = \alpha_{i-1} \circ p_i \circ p_{i+1} = 0$$

we have $\operatorname{im} \tilde{\alpha}_{i+1} \subseteq \ker b_i = \operatorname{im} b_{i+1}$ by exactness. Moreover, P_{i+1} is projective and we can lift $\tilde{\alpha}_{i+1}$ to a map $\alpha_{i+1} : P_{i+1} \to B_{i+1}$ through $\tilde{\alpha}_{i+1} = b_{i+1} \circ \alpha_{i+1}$. By definition of α_{i+1} the diagram commutes.

We proceed to show the uniqueness of $H^i(\alpha_i)$. By induction we construct maps $\gamma_i : P_i \to B_{i+1}$ such that if $(\alpha_i), (\beta_i)$ are two extensions of α we have

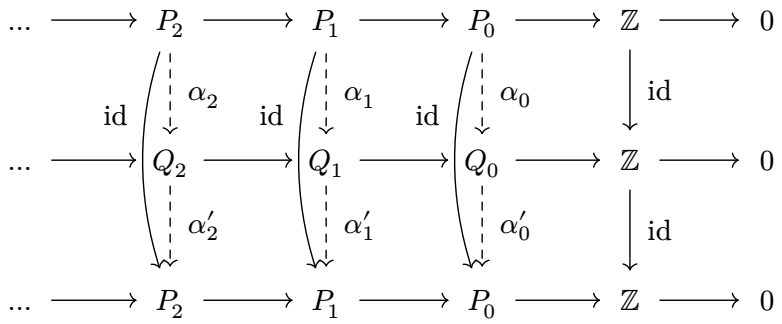
$$\alpha_i - \beta_i = \gamma_{i-1} \circ p_i + b_{i+1} \circ \gamma_i,$$

that is α_i and β_i are cochain-homotopic. This ensures they induce the same maps in cohomology. □

Proposition 2.1.6 (Good definition of group cohomology)

The cohomology groups $H^i(G, A)$ do not depend on the choice of a projective resolution $P_\bullet \to \mathbb{Z}$.

Proof: Let $P_\bullet \to A$ and $Q_\bullet \to A$ be two projective resolutions of A. Apply Lemma 2.1.4 to the following diagram:



By uniqueness of the maps in cohomology we deduce $H^i(\alpha'_i \circ \alpha_i) = \mathrm{id}$ for every i. Similarly, by swapping the roles of $P_\bullet$ and $Q_\bullet$ we have $H^i(\alpha_i \circ \alpha'_i) = \mathrm{id}$. Thus, $H^i(\alpha_i)$ and $H^i(\alpha'_i)$ are mutually inverse isomorphisms. □

Example 2.1.1 (Standard resolution)

For any $i \geq 0$ let $G^{i+1} := \underbrace{G \times ... \times G}_{i+1}$. The ring $\mathbb{Z}[G^{i+1}]$ has the structure of a G-module via

$$\sigma(\sigma_0, ..., \sigma_i) \mapsto (\sigma\sigma_0, ..., \sigma\sigma_i).$$

Moreover, this is a free G-module with basis $(1, \sigma_1, ..., \sigma_i)$. Define the maps

$$\begin{aligned} s^i_j : \mathbb{Z}[G^{i+1}] &\longrightarrow \mathbb{Z}[G^i] \\ (\sigma_0, ..., \sigma_i) &\longmapsto (\sigma_0, ..., \hat{\sigma}_j, ..., \sigma_i), \end{aligned}$$

the notation $\hat{\sigma}_j$ means the corresponding term is omitted. We use them to build G-morphisms $\delta^i : \mathbb{Z}[G^{i+1}] \to \mathbb{Z}[G^i]$ as $\delta^i = \sum_{j=0}^{i} (-1)^j s^i_j$. It is easy to check $\delta^i \circ \delta^{i+1} = 0$, thus we get a cochain complex

$$\begin{aligned} ... \xrightarrow{\delta^2} \mathbb{Z}[G^2] \xrightarrow{\delta^1} \mathbb{Z}[G^1] &\xrightarrow{\delta^0} \mathbb{Z} \to 0. \\ \sigma &\longmapsto 1 \end{aligned}$$

This is a projective (actually free!) resolution of $\mathbb{Z}$, called the *standard resolution*. To see that it is exact, fix $\sigma \in G$ and define

$$\begin{aligned} h^i : \mathbb{Z}[G^{i+1}] &\longrightarrow \mathbb{Z}[G^{i+2}], \\ (\sigma_0, ..., \sigma_i) &\longmapsto (\sigma, \sigma_0, ..., \sigma_i). \end{aligned}$$

One can check that these maps satisfy

$$\delta^{i+1} \circ h^i + h^{i-1} \circ \delta^i = \mathrm{id}_{\mathbb{Z}[G^{i+1}]}.$$

If we apply it to $x \in \ker \delta^i$ we get $x = \delta^{i+1} \circ h^i(x)$, hence $x \in \mathrm{im}\, \delta^{i+1}$. This concludes $\mathrm{im}\, \delta^{i+1} = \ker \delta^i$.

Instead of using the basis $(1, \sigma_1, ..., \sigma_i)$ of $\mathbb{Z}[G^{i+1}]$ sometimes it can be useful to consider the basis

$$[\sigma_1, ..., \sigma_i] = (1, \sigma_1, \sigma_1\sigma_2, ..., \sigma_1 ... \sigma_i).$$

With respect to this basis the boundary maps have values

$$\delta^i([\sigma_1,...,\sigma_i]) = \sigma_1[\sigma_2,...,\sigma_i] + \sum_{j=1}^{i-1}(-1)^j\big[\sigma_1,...,\sigma_j\sigma_{j+1},...\sigma_i\big] + (-1)^{i+1}[\sigma_1,...,\sigma_{i-1}].$$

We can view i-cochains as maps

$$\mathbb{Z}\big[G^{i+1}\big] \longrightarrow A,$$
$$[\sigma_1,...,\sigma_i] \longmapsto a_{\sigma_1,...,\sigma_i},$$

the inhomogeneous convention turns out to be more convenient for computations. Let's explicitate the 1-cochains: they are maps of the form $\sigma \mapsto a_\sigma$. Let ∂^i be the maps induced in cohomology by the δ^i. By definition 1-cocyles satisfy

$$0 = (\partial^2 a)[\sigma,\tau] = a(\delta^2[\sigma,\tau]) = a(\sigma[\tau]) - a([\sigma\tau]) + a([\sigma]) = \sigma a_\tau - a_{\sigma\tau} + a_\sigma.$$

On the other hand, 1-coboundaries are of the form $\partial^1\varphi$ for some $\varphi \in \mathrm{Hom}_G(\mathbb{Z}[G], A)$. If $\varphi(1) = a$ we get

$$\sigma \longmapsto \partial^1\varphi([\sigma]) = \varphi(\delta^1[\sigma]) = \varphi(1-\sigma) = a - \sigma a.$$

Thus, 1-cochains form the group

$$H^1(G,A) = \frac{\{\sigma \mapsto a_\sigma : a_{\sigma\tau} = \sigma a_\tau + a_\sigma\}}{\{\sigma \mapsto a - \sigma a\}}.$$

In particular, when A is a trivial G-group the cocycle condition becomes $a_{\sigma\tau} = a_\tau + a_\sigma$, that is $H^1(G,A) = \mathrm{Hom}(G,A)$.

Example 2.1.2 (Cohomology of cyclic groups)

Given any group G, define the *augmentation map* $\varepsilon : \mathbb{Z}[G] \to \mathbb{Z}$ by $\sigma \mapsto 1$ for every $\sigma \in G$.

Let's compute the cohomology of cyclic groups.

1. Let $G = \mathbb{Z} = \langle\sigma\rangle$, the sequence

$$0 \longrightarrow \mathbb{Z}[G] \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

   is a projective resolution. We get

$$H^0(\mathbb{Z},A) = A^{\mathbb{Z}}, \qquad H^1(\mathbb{Z},A) = \frac{A}{(\sigma-1)A}, \qquad H^i(\mathbb{Z},A) = 0 \text{ for } i \geq 2.$$

2. Let $G = C_n = \langle\sigma\rangle$ and consider the norm element $N = \sum_{i=0}^{n-1}\sigma^i$. The sequence

$$... \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

   is a projective resolution where the maps $(\sigma-1)$ and N alternate. We get

$$H^0(C_n,A) = A^{C_n}, \qquad H^{2i-1}(C_n,A) = \frac{{}_NA}{(\sigma-1)A}, \qquad H^{2i}(C_n,A) = \frac{A^{C_n}}{NA}$$

   for every $i \geq 1$, where ${}_NA := \ker N$.

2.2. Restriction, co-restriction, inflation

Definition 2.2.11 (Co-induced modules)

Let A be a G-module and let $H \leq G$. We define the *co-induced module from H to G* as

$$M_H^G(A) := \mathrm{Hom}_H(\mathbb{Z}[G], A).$$

The group G acts on $\varphi \in M_H^G(A)$ by $\sigma(\varphi)(g) = \varphi(g \cdot \sigma)$, the action is well defined: indeed, $h\varphi(g \cdot \sigma) = \varphi(hg \cdot \sigma)$ is still a H-morphism.

Lemma 2.2.5

Let A, M be G-modules and let $H \leq G$. There is a canonical isomorphism

$$\mathrm{Hom}_G\big(M, M_H^G(A)\big) \simeq \mathrm{Hom}_H(M, A).$$

Proof: Given a G-morphism

$$M \longrightarrow \mathrm{Hom}_H(\mathbb{Z}[G], A), \qquad m \longmapsto \varphi_m$$

we map it to $[m \mapsto \varphi_m(1)] \in \mathrm{Hom}_G(M, A)$. Viceversa, if $\lambda : M \to A$ is a H-morphism we map it to

$$m \longmapsto \lambda_m \in \mathrm{Hom}_H(\mathbb{Z}[G], A), \qquad \lambda_m(g) = \lambda(g \cdot m).$$

The two maps are mutually inverse isomorphisms. □

Corollary 2.2.2 (Shapiro's lemma)

There is a canonical isomorphism

$$H^i\big(G, M_H^G(A)\big) \simeq H^i(H, A).$$

Proof: Let $P_\bullet \to \mathbb{Z}$ be a projective resolution of $\mathbb{Z}$ by G-modules. Since $\mathbb{Z}[G]$ is free as a $\mathbb{Z}[H]$-module, each P_i is is also projective as a H-module. Hence, $P_\bullet \to \mathbb{Z}$ is also a projective resolution of $\mathbb{Z}$ by H-modules. By Lemma 2.2.5 we get an isomorphism of cochain complexes $\mathrm{Hom}_G\big(P_\bullet, M_H^G(A)\big) \simeq \mathrm{Hom}_H(P_\bullet, A)$ which induces the desired isomorphism in cohomology. □

Corollary 2.2.3

In particular when $H = \{1\} \leq G$ we have

$$H^i\Big(G, M_{\{1\}}^G(A)\Big) = 0 \text{ for } i > 0.$$

Proof: By Corollary 2.2.2 we have $H^i\Big(G, M_{\{1\}}^G(A)\Big) \simeq H^i(\{1\}, A)$. By using the projective resolution $0 \to \mathbb{Z} \to \mathbb{Z} \to 0$ we get that $H^i(\{1\}, A) = 0$ for $i > 0$. □

Remark 2.2.1

We call $M^G(A) := M_{\{1\}}^G(A)$ the *co-induced module from A*. There is an injective map

$$A \longrightarrow M^G(A), \qquad a \longmapsto [\sigma \mapsto \sigma a].$$

In this sense the co-induced module behaves similarly to an injective module (even if in general it is not one).

Consider now a G-module A, there is a G-morphism

$$A \xrightarrow{\sim} \mathrm{Hom}_G(\mathbb{Z}[G], A) \longrightarrow \mathrm{Hom}_H(\mathbb{Z}[G], A) = M_H^G(A)$$

corresponding to the inclusion $A \hookrightarrow M_H^G(A)$.

Definition 2.2.12 (Restriction map)

The *restriction map* is the map induced by the inclusion $A \hookrightarrow M_H^G(A)$ on the group cohomology

$$\mathrm{Res} : H^i(G, A) \longrightarrow H^i\big(G, M_H^G(A)\big) \overset{\text{Corollary 2.2.2}}{=} H^i(H, A).$$

In terms of i-cocycles it sends $\varphi : G^i \to A$ to its restriction $\varphi|_H : H^i \to A$.

Consider a normal subgroup $H \lhd G$ and a G-module A. The action of G on A^H factorizes through G/H, thus A^H is also a G/H-module. Let $P_\bullet \to \mathbb{Z}$ be a projective resolution of $\mathbb{Z}$ by G-modules and $Q_\bullet \to \mathbb{Z}$ a projective resolution of $\mathbb{Z}$ by G/H-modules. Every term Q_i inherits the structure of a G-module (although not necessarily projective), hence by Lemma 2.1.4 there is a morphism of complexes $(P_\bullet \to \mathbb{Z}) \to (Q_\bullet \to \mathbb{Z})$ obtained from the identity on $\mathbb{Z}$. Therefore we get morphisms of abelian groups

$$\mathrm{Hom}_{G/H}(Q_\bullet, A^H) = \mathrm{Hom}_G(Q_\bullet, A^H) \longrightarrow \mathrm{Hom}_G(P_\bullet, A^H),$$

they induce morphisms in cohomology

$$H^i\big(\mathrm{Hom}_{G/H}(Q_\bullet, A^H)\big) \longrightarrow H^i(\mathrm{Hom}_G(P_\bullet, A^H)) = H^i(G, A^H) \longrightarrow H^i(G, A).$$

Definition 2.2.13 (Inflation map)

The map above is called the *inflation map*:

$$\mathrm{Inf} : H^i(G/H, A^H) \longrightarrow H^i(G, A).$$

In terms of i-cocycles it sends $\varphi : (G/H)^i \to A^H$ to $\varphi \circ \pi_H : G^i \to (G/H)^i \to A^H \subseteq A$.

Lemma 2.2.6

1. We have $M^G(A)^H \simeq M^{G/H}(A)$ (in particular $H^i\big(G/H, M^G(A)^H\big) = 0$ for $i > 0$).
2. We have $H^i\big(H, M^G(A)\big) = 0$ for $i > 0$.

Proof:

1. By definition $M^G(A)^H = \mathrm{Hom}(\mathbb{Z}[G], A)^H$. Given a morphism $\varphi : G \to A$, it being H-invariant means $\varphi(\sigma h) = h\varphi(\sigma) = \varphi(\sigma)$. Therefore it factors through $\varphi : G \to G/H \to A$ and

$$\mathrm{Hom}(\mathbb{Z}[G], A)^H \simeq \mathrm{Hom}(\mathbb{Z}[G/H], A).$$

2. We know $\mathbb{Z}[G]$ is a free $\mathbb{Z}[H]$-module, therefore $M^G(A) = \prod_{i\in I} M^H(A)$. For every $i > 0$ we get

$$H^i\big(H, M^G(A)\big) \simeq H^i\left(H, \prod_{i\in I} M^H(A)\right) \simeq \prod_{i\in I} H^i(H, M^H(A)) = 0.$$

□

Proposition 2.2.7 (Restriction-inflation sequence)

1. There is an exact sequence
$$0 \longrightarrow H^1(G/H, A^H) \xrightarrow{\text{Inf}} H^1(G, A) \xrightarrow{\text{Res}} H^1(H, A).$$
2. If moreover $H^1(H, A) = 0$ there is an exact sequence
$$0 \longrightarrow H^2(G/H, A^H) \xrightarrow{\text{Inf}} H^2(G, A) \xrightarrow{\text{Res}} H^2(H, A).$$

Proof:

1. *Exercise.*
2. Let $C_A = M^G(A)/A$, there is an exact sequence
$$0 \longrightarrow A \longrightarrow M^G(A) \longrightarrow C_A \longrightarrow 0.$$
From the group cohomology long exact sequence we extract the short exact sequence
$$0 \longrightarrow A^H \longrightarrow M^G(A)^H \longrightarrow C_A^H \longrightarrow H^1(H, A) = 0.$$
We construct a commutative diagram in the following way: we take the long exact sequences in group cohomology extracted from the above short sequences for columns, the restriction and inflation maps for rows:

$$\begin{array}{ccccccc}
 & & H^1\big(G/H, M^G(A)^H\big) & & H^1\big(G, M^G(A)\big) & & H^1\big(H, M^G(A)\big) \\
 & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & H^1\big(G/H, C_A^H\big) & \xrightarrow{\text{Inf}} & H^1(G, C_A) & \xrightarrow{\text{Res}} & H^1(H, C_A) \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & H^2(G/H, A^H) & \xrightarrow{\text{Inf}} & H^2(G, A) & \xrightarrow{\text{Res}} & H^2(H, A) \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & H^2\big(G/H, M^G(A)^H\big) & & H^2\big(G, M^G(A)\big) & & H^2\big(H, M^G(A)\big)
\end{array}$$

We shall prove the central vertical maps are isomorphisms, then by the previous point we get the desired result. This is a direct consequence of Corollary 2.2.3 and Lemma 2.2.6.

□

Remark 2.2.2

In general, there is a spectral sequence $H^p(G/H, H^q(H, A)) \Rightarrow H^{p+q}$ (G,A) called the *Hochschild-Serre spectral sequence*.

Let $H \leq G$ be a subgroup of finite index $n = [G : H]$ and fix $\rho_1, ..., \rho_n$ a system of coset representatives of $G \bmod H$. Identify $A \simeq \mathrm{Hom}_G(\mathbb{Z}[G], A)$ and consider the map

$$M_H^G(A) = \mathrm{Hom}_H(\mathbb{Z}[G], A) \longrightarrow \mathrm{Hom}_G(\mathbb{Z}[G], A) \simeq A,$$
$$\varphi \longmapsto \left[\varphi_H^G : x \mapsto \sum_{j=1}^{n} \rho_j \varphi\big(\rho_j^{-1} x\big)\right].$$

It is well defined, i.e. it does not depend on the choice of $\rho_1, ..., \rho_n$. Let $\rho_j \tau_j$ be another set of respresentatives, we can assume $\tau_j \in H$. We have

$$\rho_j \tau_j \varphi\big(\tau_j^{-1} \rho_j^{-1} x\big) = \rho_j \tau_j \tau_j^{-1} \varphi\big(\rho_j^{-1} x\big) = \rho_j \varphi\big(\rho_j^{-1} x\big),$$

since φ is a H-morphism. Moreover, the map φ_H^G is effectively a G-morphism, indeed

$$\varphi_H^G(\sigma x) = \sum_j \rho_j \varphi\big(\rho_j^{-1} \sigma x\big) = \sigma \sum_j \big(\sigma^{-1} \rho_j\big) \varphi\Big(\big(\sigma^{-1} \rho_j\big)^{-1} x\Big) = \sigma \varphi_H^G(x)$$

because $\sigma^{-1} \rho_j$ is still a set of representatives.

Definition 2.2.14 (Co-restriction map)

The *co-restriction map* is the morphism

$$\mathrm{Cor} : H^i(H, A) \simeq H^i\big(G, M_H^G(A)\big) \longrightarrow H^i(G, A)$$

induced in cohomology by the map above.

Proposition 2.2.8

If $[G : H] = n$ then $\mathrm{Cor} \circ \mathrm{Res} = n$.

Proof: Recall $A \simeq \mathrm{Hom}_G(\mathbb{Z}[G], A)$. The maps that induce restriction and co-restriction in cohomology compose as

$$\begin{array}{ccccc} \mathrm{Hom}_G(\mathbb{Z}[G], A) & \longrightarrow & \mathrm{Hom}_H(\mathbb{Z}[G], A) & \longrightarrow & \mathrm{Hom}_G(\mathbb{Z}[G], A), \\ \varphi & \longmapsto & \varphi & \longmapsto & \varphi_H^G. \end{array}$$

Since φ is a G-morphism we have

$$\varphi_H^G(x) = \sum_{j=1}^{n} \rho_j \varphi\big(\rho_j^{-1} x\big) = \sum_{j=1}^{n} \rho_j \rho_j^{-1} \varphi(x) = \sum_{j=1}^{n} \varphi(x) = n\varphi(x),$$

therefore the map $\varphi \mapsto \varphi_H^G$ is multiplication-by-n. □

Corollary 2.2.4

If G is finite of order n, then $n \cdot H^i(G, A) = 0$ for every $i > 0$.

Proof: We apply Res and Cor with subgroup $H = \{1\} \leq G$. We get

$$H^i(G,A) \xrightarrow{\text{Res}} H^i(\{1\},A) \xrightarrow{\text{Cor}} H^i(G,A),$$

from Proposition 2.2.8 this is the multiplication-by-n map. On the other hand,

$$H^i(\{1\},A) \overset{\text{Corollary 2.2.2}}{\simeq} H^i\big(G,M^G(A)\big) \overset{\text{Corollary 2.2.3}}{=} 0 \text{ for } i>0.$$

Hence $\cdot n = 0$ as maps on $H^i(G,A)$. □

Corollary 2.2.5

If G has finite order n and A is a $\mathbb{Q}$-vector space, then $H^i(G,A)=0$ for every $i>0$.

Proof: The multiplication map $a \mapsto na$ is an isomorphism $A \xrightarrow{\sim} A$. The induced map $H^i(G,A) \xrightarrow{n} H^i(G,A)$ is also an isomorphism, and Corollary 2.2.4 yields $H^i(G,A)=0$. □

The proof can be adapted to any K-vector space where K is n-divisible, that is multiplication by n is surjective on K.

2.3. Profinite groups

Definition 2.3.15 (Inverse system of groups)

Definition 2.3.16 (Profinite group)

Example 2.3.3

Example 2.3.4

Definition 2.3.17 (Direct system of groups)

Definition 2.3.18 (Continuous G-modules)

Definition 2.3.19 (Continuous cohomology of groups)

Example 2.3.5

2.4. Cup product

3. p-adic number fields

Definition 3.0.20 (Brauer group)

Let k be a field and k_s be one of its separable closures. The *Brauer group* of k is

$$\mathrm{Br}(k) := H^2(k, k_s)$$

If $K|k$ is a separable subextension of $k_s|k$ we can define the *relative Brauer group* of the extension as

$$\mathrm{Br}(K|k) := \ker\left(\mathrm{Br}(k) \xrightarrow{\mathrm{Res}} \mathrm{Br}(K)\right)$$

The elements of $\mathrm{Br}(k)$ correspond to the finite dimensional division algebras D with center k. In this correspondence, the subgroup $\mathrm{Br}(K|k)$ gives those D for which $D \otimes_k K \simeq M_{n\times n}(K)$, where $n^2 = \dim_k D$.

Lemma 3.0.7

1. If $K|k$ is a Galois extension with group G then it holds $\mathrm{Br}(K|k) \simeq H^2(G, K^\times)$;
2. The Brauer group $\mathrm{Br}(k)$ is the union of all the subgroups $\mathrm{Br}(K|k)$ for finite Galois extensions $K|k$;
3. If G is finite and cyclic it holds $\mathrm{Br}(K|k) \simeq \frac{k^\times}{N_{K|k}(K^\times)}$.

Here $N_{K|k} : K^\times \to k^\times$ denotes the norm of the field extension. When $K|k$ is finite and Galois it has value

$$N_{K|k}(a) = \prod_{\sigma\in \mathrm{Gal}(K|k)} \sigma a$$

for every $a \in K^\times$.

Proof:

1. From Hilbert's theorem 90 we have $H^1(G, K^\times) = 0$. As seen in Proposition 2.2.7 this makes the following inflation-restriction sequence exact:

   $$0 \longrightarrow H^2(G, K^\times) \xrightarrow{\mathrm{Inf}} H^2(k, k_s) \xrightarrow{\mathrm{Res}} H^2(K, k_s),$$

   and by definition $\mathrm{Br}(K|k) = \ker \mathrm{Res} = H^2(G, K^\times)$.
2. From the previous point we deduce

   $$\mathrm{Br}(k) = \varinjlim H^2(G, K^\times) \overset{(1)}{=} \varinjlim \mathrm{Br}(K|k) = \bigcup \mathrm{Br}(K|k).$$
3. As seen in Example 2.1.2 when the Galois group G is cyclic we have

   $$\mathrm{Br}(K|k) \overset{(1)}{=} H^2(G, K^\times) = \frac{(K^\times)^G}{N_{K|k}(K^\times)} = \frac{k^\times}{N_{K|k}(K^\times)}.$$

□

Example 3.0.6

0. When the field k is complete we have $\mathrm{Br}(k) = 0$ since the Galois action on it is trivial.

1. Let's compute the Brauer group of the real numbers:

$$\mathrm{Br}(\mathbb{R}) = \mathrm{Br}(\mathbb{C}|\mathbb{R}) = H^2(\mathrm{Gal}(\mathbb{C}|\mathbb{R}), \mathbb{C}^\times) = \frac{\mathbb{R}^\times}{N_{\mathbb{C}|\mathbb{R}}(\mathbb{C}^\times)} = \frac{\mathbb{R}^\times}{\mathbb{R}^\times_{>0}} \simeq \mathbb{Z}/2\mathbb{Z}.$$

2. When $k = \mathbb{F}_q$ is a finite field the Brauer group vanishes: $\mathrm{Br}(\mathbb{F}_q) = 0$. By Lemma 3.0.7 it is enough to show that $\mathrm{Br}(\mathbb{F}_{q^r}|\mathbb{F}_q) = 0$ for every $r > 0$. Recall the following facts:
   - The group $\mathrm{Gal}(\mathbb{F}_{q^r}|\mathbb{F}_q)$ is cyclic, generated by the frobenius $a \mapsto a^q$. In particular the norm has value
   $$N(a) = a^{\frac{q^r-1}{q-1}};$$
   - The multiplicative group $\mathbb{F}_{q^r}^\times$ is cyclic.

   Let ω be a generator of $\mathbb{F}_{q^r}^\times$, its norm $N(\omega)$ by the above calculation has order $q-1$ and therefore generates $\mathbb{F}_q$. This shows N is surjective and from Lemma 3.0.7 we deduce $\mathrm{Br}(\mathbb{F}_{q^r}|\mathbb{F}_q) = 0$.

3.1. The path to $K|\mathbb{Q}_p$

The main goal of this section is proving that if $K|\mathbb{Q}_p$ is a p-adic number field then its Brauer group is $\mathrm{Br}(K) = \mathbb{Q}/\mathbb{Z}$.

Definition 3.1.21 (Discretely valuated fields)

A *discretely valuated field* is a field equipped with a discrete valuation $v : K \to \mathbb{Z} \cup \{\infty\}$, i.e. it holds
- $v(x) = \infty \iff x = 0$;
- $v(xy) = v(x)v(y)$;
- $v(x+y) \geq \min\{v(x), v(y)\}$.

Moreover we say K is *complete* if it complete with respect to the norm $\|x\|_v = e^{-v(x)}$. The *valuation ring* of K is defined as

$$\mathcal{O}_K := \{x \in K : v(x) \geq 0\},$$

it is a discrete valuation ring with maximal ideal $M_K := \{x \in K : v(x) > 0\}$. The *residue field* of K is the quotient $k(v) := \mathcal{O}_K/M_K$.

Fact

The ideal M_k is principal and it is generated by any element $\pi \in K$ with valuation $v(\pi) = 1$, such an element is called a *uniformiser* for $\mathcal{O}_K$. Since M_K is maximal, $k(v) = \mathcal{O}_K/(\pi)$ is indeed a field.

Fact

The field K is complete if and only if its valuation ring $\mathcal{O}_K$ is a complete DVR, that is

$$\mathcal{O}_K \simeq \varprojlim_i \frac{\mathcal{O}_K}{M_K^i}.$$

Example 3.1.7

1. The p-adic field $\mathbb{Q}_p$ is complete as $\mathbb{Q}_p = \mathrm{Frac}(\lim_{\leftarrow} \mathbb{Z}/p^i\mathbb{Z})$.
2. The Laurent series $k((t))$ form a complete field as $k((t)) = \mathrm{Frac}(\lim_{\leftarrow} k[\![t]\!]/(t^i))$.

Fact

If (K, v) is a complete discretely valuated field and $L|K$ is a finite separable extension, the valuation can be extended uniquely to L via

$$w = \frac{1}{[L:K]} v \circ N_{L|K} : L^\times \to \frac{1}{[L:K]}\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}.$$

The discretely valuated field (L, w) is complete itself and its valuation ring $\mathcal{O}_L$ is the integral closure of $\mathcal{O}_K$ inside L.

Definition 3.1.22 (ramification and residue class)

If $L|K$ is a finite separable extension of complete discretely valuated fields, the associated *ramification index* is $e(L|K) := [w(L^\times) : v(K^\times)]$ and the *residue class degree* is $f(L|K) := [k(w) : k(v)]$.

We say $L|K$ is *unramified* if $k(w)|k(v)$ is separable and $e = 1$; and $L|K$ is *totally ramified* if $f = 1$.

From now on we assume that K is a complete discretely valuated field.

Facts

- When $k(w)|k(v)$ is separable it holds $[L:K] = e(L|K) \cdot f(L|K)$.
- If $L|K$ is finite and unramified there exists a unique extension $L \supset M \supset K$ such that $M|K$ is unramified and $L|M$ is totally ramified.
- *Towers property:* If both $L|K, L'|L$ are finite and unramified, $L'|K$ is also finite and unramified.
- *Shift property:* If $L|K$ is finite and unramified, $K'|K$ is finite and separable; the extension $LK'|K'$ is finite and unramified.

Definition 3.1.23 (Unramified closure)

We set the maximal unramified extension of K as

$$K_{\mathrm{nr}} = \bigcup \{\overline{K} \supset L \supset K : L|K \text{ is finite and unramified}\}.$$

The maximal unramified extension K_{nr} is a dicretely valuated field, with valuation extending that of K. Its residue field is the separable closure $k(v)_s$, however in general it is not a complete field.

Suppose $L|K$ is a finite Galois extension (this can be generalised to infinite Galois extensions) and let $G = \mathrm{Gal}(L|K)$. The action of G on L is compatible with the valuation (this

is a consequence of the uniqueness of valuation extensions), meaning $\mathcal{O}_L$ and M_L are G -invariant. Every element $\sigma \in G$ induces an automorphism

$$\overline{\sigma} : \frac{\mathcal{O}_L}{M_L} \to \frac{\mathcal{O}_K}{M_K}, \qquad \overline{x} \mapsto \overline{\sigma(x)},$$

giving a homomorphism on the Galois group of the residue fields extension

$$\rho : \mathrm{Gal}(L|K) \to \mathrm{Gal}(k(w)|k(v)).$$

By Proposition 1.1.2 this is surjective, moreover its kernel is given by the *inertia subgroup* $I(L|K)$ and it is known that $|I(L|K)| = e(L|K)$. In particular when the extension is unramified (i.e. $e = 1$) the map ρ is an isomorphism.

Notice that by a similar argument if $L|K$ is a finite unramified extension then $L|K$ is Galois if and only if the residue field extension $k(w)|k(v)$ is Galois.

Example 3.1.8

Suppose the residue field $k(v)$ is finite. Since finite extension on finite fields are Galois, every finite unramified extension $L|K$ is a cyclic Galois extension. This means that for every $n \mid [L : K]$ there is a unique unramified subextension $L \supset K_n \supset K$ of degree $[K_n : K] = n$. In other words if we fix a closure $\overline{K}$ for every n there is a unique unramified Galois extension $\overline{K} \supset K_n \supset K$ of degree n.

Definition 3.1.24 (Units and principal units groups)

The *units group* of K is the subgroup $\mathcal{U}_K := \{a \in K^\times : v(a) = 0\}$. The *$i$-th group of principal units* is $\mathcal{U}_K^{(i)} := \{a \in K^\times : v(a-1) \geq i\}$.

There is an exact sequence of groups

$$1 \longrightarrow \mathcal{U}_K \longrightarrow K^\times \xrightarrow{v} \mathbb{Z} \longrightarrow 1,$$

the units group can be interpreted as the kernel of the valuation map. Moreover there is a surjection $\mathcal{U}_K \twoheadrightarrow k(v)^\times$ given by $u \mapsto u \bmod M_K$ and the principal units group can be viewed as its kernel:

$$\mathcal{U}_K^{(1)} = \ker(\mathcal{U}_K \longrightarrow k(v)^\times).$$

Notice that $\bigcap_i \mathcal{U}_K^i = \{1 + a \in K^\times : v(a) \geq i \text{ for each } i \geq 1\} = \{1\}$. The map $\mathcal{U}_K^{(i)} \to M_K^i$ given by $u \mapsto u \bmod M_K^{i+1}$ induces an isomorphism

$$\frac{\mathcal{U}_K^{(i)}}{\mathcal{U}_K^{(i+1)}} \xrightarrow{\sim} \frac{M_K^i}{M_K^{i+1}} \simeq \frac{\pi^i \mathcal{O}_K}{\pi^{i+1}\mathcal{O}_K} \simeq \frac{\mathcal{O}_K}{(\pi)} \simeq k(v).$$

We have therefore constructed a filtration (consisting of an infinite number of elements)

$$\mathcal{U}_K \underset{k(v)^\times}{\supset} \mathcal{U}_K^{(1)} \underset{k(v)^+}{\supset} \mathcal{U}_K^{(2)} \underset{k(v)^+}{\supset} \ldots \supset \{1\},$$

this filtration can be useful to prove properties of the multiplicative group $K^\times$.

Example 3.1.9

Consider a finite p-adic extension $K|\mathbb{Q}_p$. The residue field $k(v)$ is a finite extension of $\mathbb{F}_p$, hence it is of the form $k(v) = \mathbb{F}_q$ where $q = p^r, r = [K : \mathbb{Q}_p]$. The exact sequence

$$1 \longrightarrow \mathcal{U}_K^{(1)} \longrightarrow \mathcal{U}_K \longrightarrow k(v)^\times \to 1$$

splits by lifting $k(v)^\times$ to the $q-1$ roots of unity in $\mathcal{U}_K$ via $k(v)^\times \to \mu_{q-1}$. This gives the decomposition

$$K^\times \simeq \mathbb{Z} \times \mathcal{U}_K \simeq \mathbb{Z} \times \mu_{q-1} \times \mathcal{U}_K^{(1)}.$$

It can be shown that $\mathcal{U}_K^{(1)}$ itself can be decomposed as $\mathcal{U}_K^{(1)} \simeq \mu_{p^s} \times \mathbb{Z}_p^r$, where p^s is the number of p-power roots of unity in K.

We deduce $K^\times / K^{\times n}$ is a finite abelian group.

3.2. The Hasse invariant

The goal of this section is to prove the Hasse theorem on Brauer groups of complete discretely valuated fields with finite residue field. In particular, this applies to finite p-adic extensions $K|\mathbb{Q}_p$. We then proceed to use this result to define the Hasse invariant map, that underlies the generalization of the Hilbert symbol. to be updated.

The main result is that in the discussed hypothesis $\mathrm{Br}(K) \simeq \mathbb{Q}/\mathbb{Z}$. We divide the proof in two steps: we first prove $\mathrm{Br}(K_{\mathrm{nr}}|K) \simeq \mathbb{Q}/\mathbb{Z}$, secondly we show $\mathrm{Br}(K_{\mathrm{nr}}|K) = \mathrm{Br}(K)$.

From now on let K be a complete discretely valued field with finite residue field κ.

3.2.1. Showing $\mathrm{Br}(K_{\mathrm{nr}}|K) \simeq \mathbb{Q}/\mathbb{Z}$

Lemma 3.2.8

Let $L|K$ be a finite unramified Galois extension and let $G = \mathrm{Gal}(L|K)$. There is a split short exact sequence

$$0 \longrightarrow H^2(G, \mathcal{U}_L) \longrightarrow H^2(G, L^\times) \longrightarrow \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) \longrightarrow 0.$$

Proof: The short exact sequence

$$1 \longrightarrow \mathcal{U}_L \longrightarrow L^\times \xrightarrow{v} \mathbb{Z} \longrightarrow 0$$

splits via $s : \mathbb{Z} \to L^\times, 1 \mapsto \pi$ where π is a uniformiser of $\mathcal{O}_L$. Since the extension is unramified, we can choose $\pi \in K$, so that the section s is G-equivariant. Thus s induces a map on cohomology (that we still denote s) and the exact sequence

$$0 = H^1(G, \mathbb{Z}) \longrightarrow H^2(G, \mathcal{U}_L) \longrightarrow H^2(G, L^\times) \longrightarrow H^2(G, \mathbb{Z}) \longrightarrow 0$$

also splits via the section $s : H^2(G, \mathbb{Z}) \to H^2(G, L^\times)$ (indeed, the existence of the section guarantees surjectivity of the right-hand map). It remains to justify that $H^1(G, \mathbb{Z}) = 0$: since the G-action on $\mathbb{Z}$ is trivial we have $H^1(G, \mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Z})$ and every homomorphism from a finite group to $\mathbb{Z}$ is trivial.

It remains to identify $H^2(G, \mathbb{Z})$ with $\mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z})$. Consider the short exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0,$$

the associated long sequence in cohomology gives

$$0 = H^1(G, \mathbb{Q}) \longrightarrow H^1(G, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\sim} H^2(G, \mathbb{Z}) \longrightarrow H^2(G, \mathbb{Q}) = 0.$$

From Corollary 2.2.5 the cohomology of $\mathbb{Q}$ vanishes in positive dimension. Since the G-action on $\mathbb{Q}/\mathbb{Z}$ is trivial we have $H^1(G, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z})$ and we conclude. □

Lemma 3.2.9

Let G be a finite group and let (A_j, φ_j) be an inverse system of G-modules indexed by $\mathbb{Z}_{>0}$ (in particular the transition maps are the $\varphi_j : A_{j+1} \to A_j$). For any i, if $H^i(G, A_1) = 0$ and $H^i(G, \ker \varphi_j) = 0$ for every j, then $H^i\left(G, \lim_{\leftarrow} A_j\right) = 0$.

Lemma 3.2.10 (Additive Hilbert's 90)

If K is a field and $L|K$ is a finite Galois extension with group G, then $H^i(G, L^+) = 0$ for every $i > 0$.

In particular from Lemma 3.2.9 we deduce $H^i(k, k_s) = 0$ for every $i > 0$.

Proof: Recall the *normal basis theorem*: there is an element $a \in L$ such that the set $\{\sigma a : a \in G\}$ forms a k-basis for L. We deduce there is an isomorphism of G-modules

$$k^+ \otimes_{\mathbb{Z}} \mathbb{Z}[G] \to L^+, \qquad \lambda_i \otimes \sigma_i \mapsto \lambda_i \sigma_{i(a)}.$$

Moreover there is a non-canonical isomorphism $k^+ \otimes_{\mathbb{Z}} \mathbb{Z}[G] \simeq \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], k^+)$ of k-vector spaces since they have the same (finite) dimension. The latter is the coinduced-G-module $M^G(k^+)$ and therefore by Corollary 2.2.2 we conclude $H^i\left(G, M^G(k^+)\right) = 0$. □

Proposition 3.2.9

Let $L|K$ be a finite unramified Galois extension with Galois group G. Let κ and λ denote the residue fields of K and L, respectively. The reduction map $\mathcal{U}_L \to \lambda^\times$ induces an isomorphism $H^2(G, \mathcal{U}_L) \xrightarrow{\sim} H^2(G, \lambda^\times)$.

Proof: The quotients $\mathcal{U}_L^{(1)}/\mathcal{U}_L^{(j)}$ form an inverse system of groups and since the principal units groups $\mathcal{U}_L^{(j)}$ have trivial intersection we get an isomorphism $\mathcal{U}_L^{(1)} \simeq \lim_{\leftarrow} \mathcal{U}_L^{(1)}/\mathcal{U}_L^{(j)}$. For every $j \geq 1$ there is an exact sequence

$$1 \longrightarrow \mathcal{U}_L^{j+1} \longrightarrow \mathcal{U}_L^{j} \longrightarrow \lambda^+ \longrightarrow 1 \Longrightarrow \lambda^+ \simeq \mathcal{U}_L^{(j)}/\mathcal{U}_L^{(j+1)}.$$

For every $i > 0$, by Lemma 3.2.10 we have $H^i(G, \lambda^+) = 0$ and therefore $H^i\left(G, \mathcal{U}_L^{(j)}/\mathcal{U}_L^{(j+1)}\right) = 0$. Applying Lemma 3.2.9 yields $H^i\left(G, \mathcal{U}_L^{(1)}\right) = 0$. Now extract the cohomology long exact sequence from $1 \to \mathcal{U}_L^{(1)} \to \mathcal{U}_L \to \lambda^\times \to 1$, we get

$$0 = H^2\left(G, \mathcal{U}_L^{(1)}\right) \longrightarrow H^2(G, \mathcal{U}_L) \xrightarrow{\sim} H^2(G, \lambda^\times) \longrightarrow H^3\left(G, \mathcal{U}_L^{(1)}\right) = 0,$$

that is the isomorphism we desired. □

Corollary 3.2.6

We have a split short exact sequence

$$0 \longrightarrow \mathrm{Br}(\lambda|\kappa) \longrightarrow \mathrm{Br}(L|K) \longrightarrow \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) \longrightarrow 0.$$

In particular when κ is finite, if we let $n = [L : K]$, then $\mathrm{Br}(L|K) \simeq \mathbb{Z}/n\mathbb{Z}$.

Proof: By Lemma 3.2.8 and Proposition 3.2.9 we get the desired sequence. When κ is finite, since the extension $\lambda|\kappa$ is itself finite (recall that $\mathrm{Gal}(\lambda|\kappa) \simeq \mathrm{Gal}(L|K)$ in our case) we have $\mathrm{Br}(\lambda|\kappa) = 0$. Moreover $n = [L : K] = |G|$ and therefore $\mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$. □

We can now apply the direct limit over all finite unramified Galois extension $L|K$ to the short exact sequence and get the following result:

Theorem 3.2.3 ($\mathrm{Br}(K_{\mathrm{nr}}|K) \simeq \mathbb{Q}/\mathbb{Z}$)

If K is a complete dicretely valuated field with perfect residue field κ, then we have a split short exact sequence

$$0 \longrightarrow \mathrm{Br}(\kappa) \longrightarrow \mathrm{Br}(K_{\mathrm{nr}}|K) \longrightarrow \mathrm{Hom}_{\mathrm{cont}}(\mathrm{Gal}(\overline{\kappa}|\kappa), \mathbb{Q}/\mathbb{Z}) \longrightarrow 0.$$

If moreover κ is finite we get $\mathrm{Br}(K_{\mathrm{nr}}|K) \simeq \mathbb{Q}/\mathbb{Z}$.

3.2.2. Showing $\mathrm{Br}(K_{\mathrm{nr}}|K) \simeq \mathrm{Br}(K)$

This fact is in general true for any complete discretely valuated field, however we are only going to prove it for fields K with a finite residue field.

Lemma 3.2.11

If K is a complete discretely valuated field and $L|K$ is a finite Galois extension with Galois group G, then the units group $\mathcal{U}_L$ contains a G-submodule $V_L^1 \subset \mathcal{U}_L$ of finite index such that $H^i(G, V_L^1) = 0$ for every $i > 0$.

Proof: From the *normal basis theorem* there is an $a \in L$ such that the set $\{\sigma a : \sigma \in G\}$ is a k-basis of L. Let π be a uniformiser of $\mathcal{O}_K$ (i.e. $v_K(\pi) = 1$), since G is finite there is a $r \gg 0$ such that $\pi^r \sigma a \in \mathcal{O}_L$ for every $\sigma \in G$. Define the $\mathcal{O}_K$-module $M = \langle \pi^r \sigma a : \sigma \in G \rangle_{\mathcal{O}_K}$, it is a submodule of $\mathcal{O}_L$. We know $\mathcal{O}_L$ is free of rank $[L : K]$ and by construction this also holds for M. By the structure theorem of finitely generated modules on PIDs the quotient $\mathcal{O}_L/M$ is finite (since it is torsion and finitely generated). This shows that $\pi^m \mathcal{O}_L \subset M$ for a suitable m, fix such an m. Define $V_L^j := 1 + \pi^{m+j} M \subset \mathcal{U}_L$ and consider the map

$$V_L^j \to \frac{M}{\pi M}, \qquad 1 + \pi^{m+j} a \mapsto a \bmod \pi M.$$

Its kernel is exactly V_L^{j+1}, therefore the map induces an isomorphism

$$\frac{V_L^j}{V_L^{j+1}} \xrightarrow{\sim} \frac{M}{\pi M} \simeq \frac{\mathcal{O}_K}{(\pi)} \otimes \mathbb{Z}[G] \simeq \kappa \otimes \mathbb{Z}[G].$$

By Corollary 2.2.2, the cohomology group $H^i(G, \kappa \otimes \mathbb{Z}[G]) = 0$ since $\kappa \otimes \mathbb{Z}[G]$ is isomorphic to a coinduced-module. Moreover $V_L^1 \simeq \varprojlim V_L^1/V_L^j$ and by Lemma 3.2.9 we get $H^1(G, V_L^1) = 0$.

Finally, $V_L^1 \subset \mathcal{U}_L$ has finite index since it contains the subgroup $1+\pi^{2m+1}\mathcal{O}_L$, which has finite index in $\mathcal{U}_L$. □

Corollary 3.2.7

The cohomology group $H^i(G,\mathcal{U}_L)$ is finite for every $i>0$.

Proof: By the short exact sequence $1 \to V_L^1 \to \mathcal{U}_L \to \mathcal{U}_L/V_L^1 \to 1$ we extract the cohomology long exact sequence

$$0 = H^i(G,V_L^1) \longrightarrow H^i(G,\mathcal{U}_L) \xrightarrow{\sim} H^i(G,\mathcal{U}_L/V_L^1) \longrightarrow H^{i+1}(G,V_L^1) = 0,$$

Indeed, by the previous lemma $H^i(G,V_L^1)=0$ and the quotient $\mathcal{U}_L/V_L^1$ is finite. □

Lemma 3.2.12

Let G be a finite cyclic group and let A be a finite G-module. We have

$$\big|H^1(G,A)\big| = \big|H^2(G,A)\big|.$$

Proof: We have computed the cohomology of a cyclic group in Example 2.1.2:

$$H^1(G,A) = \frac{{}_NA}{(\sigma-1)A}, \qquad H^2(G,A) = \frac{A^G}{NA},$$

where σ is a generator of G and $N=\sum_{\sigma\in G}\sigma$. We have two short exact sequences

$$0 \longrightarrow A^G \longrightarrow A \longrightarrow (\sigma-1)A \longrightarrow 0$$
$$0 \longrightarrow {}_NA \longrightarrow A \longrightarrow NA \longrightarrow 0$$

from which we deduce $\big|A^G\big||(\sigma-1)A| = |A| = |{}_NA||NA|$. □

Proposition 3.2.10

If K is a complete discretely valuated field and $L|K$ is a finite Galois extension with cyclic Galois group $G \simeq \mathbb{Z}/n\mathbb{Z}$, then the relative Brauer group has order $|\mathrm{Br}(L|K)| = n$.

Proof: We need to show that $H^2(G,L^\times)$ is finite of order n. From the short exact sequence $1 \to \mathcal{U}_L \to L^\times \to \mathbb{Z} \to 1$ we extract the cohomology long exact sequence

$$0 = H^1(G,\mathbb{Z}) \longrightarrow H^2(G,\mathcal{U}_L) \longrightarrow H^2(G,L^\times) \longrightarrow H^2(G,\mathbb{Z}) \longrightarrow H^3(G,\mathcal{U}_L) \longrightarrow H^3(G,L^\times) = 0.$$

Indeed since G is finite $H^1(G,\mathbb{Z})=0$ and since G is also cyclic $H^3(G,L^\times) = H^1(G,L^\times) = 0$ by Hilbert's 90. Since all the elements in the sequence have finite order we deduce

$$\big|H^2(G,L^\times)\big|\big|H^3(G,\mathcal{U}_L)\big| = \big|H^2(G,\mathcal{U}_L)\big|\big|H^2(G,\mathbb{Z})\big| \overset{\text{Lemma 3.2.12}}{\Longrightarrow} \big|H^2(G,L^\times)\big| = \big|H^2(G,\mathbb{Z})\big|.$$

Since $H^2(G,\mathbb{Z}) \simeq \mathrm{Hom}(G,\mathbb{Q}/\mathbb{Z}) \simeq \mathbb{Z}/n\mathbb{Z}$ we conclude. □

Proposition 3.2.11

If $L|K$ is a finite separable extension the following diagram commutes:

diagram

Proof: Let $L \supset M \supset K$ be the unique middle extension such that $L|M$ is totally ramified and $M|K$ is unramified. We prove the theorem separatedly for totally ramified and unramified extensions, then since both Res and the rank satisfy the tower property we conclude it must hold for all $L|K$.

In both cases we are going to use the following fact: if G is a profinite group, the long group cohomology sequence extracted from $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ gives

$$0 = H^1(G, \mathbb{Q}) \longrightarrow H^1(G, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\sim} H^2(G, \mathbb{Z}) \longrightarrow H^2(G, \mathbb{Q}) = 0.$$

The isomorphism follows from Corollary 2.2.5

- $\boxed{L|K \text{ is unramified}}$ In particular we have $L_{\mathrm{nr}} = K_{\mathrm{nr}}$. Since the restriction map Res is functorial (*Exercise.*) the following diagram commutes:

  $$\mathrm{Br}(K_{\mathrm{nr}}|K) = H^2(\mathrm{Gal}(K_{\mathrm{nr}}|K), K_{\mathrm{nr}}^\times) \xrightarrow{v} H^2(\mathrm{Gal}(K_{\mathrm{nr}}|K), \mathbb{Z}) \simeq H^2(\mathrm{Gal}(\overline{\kappa}|\kappa), \mathbb{Z}) \simeq H^1(\kappa, \mathbb{Q}/\mathbb{Z})$$

  diagram

  ^here $H^1(\kappa, \mathbb{Q}/\mathbb{Z}) = H^1(\mathrm{Gal}(\overline{\kappa}|\kappa), \mathbb{Q}/\mathbb{Z})$, right? Moreover, since $\mathbb{Q}/\mathbb{Z}$ is a trivial module we get

  $$H^1(\kappa, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}(\kappa, \mathbb{Q}/\mathbb{Z}) \simeq \mathrm{Hom}\left(\hat{\mathbb{Z}}, \mathbb{Q}/\mathbb{Z}\right) \simeq \mathbb{Q}/\mathbb{Z}.$$

  The isomorphism $\mathrm{Gal}(\overline{\kappa}|\kappa) \simeq \hat{\mathbb{Z}}$ is obtained by sending 1 to the Frobenius map $x \mapsto x^{|\kappa|}$. Analagously $\mathrm{Gal}(\overline{\kappa}|\lambda) \simeq \hat{\mathbb{Z}}$ sends 1 to $x \mapsto x^{|\lambda|}$, hence the restriction map under this isomorphism is

  $$\left[x \mapsto x^{|\kappa|}\right] \longmapsto \left[x \mapsto x^{|\lambda|} = x^{|\kappa|\,[\lambda:\kappa]}\right],$$

  which becomes multiplication by $[\lambda : \kappa]$ on $\mathbb{Q}/\mathbb{Z}$. Since the extension is unramified (is this why?) $[\lambda : \kappa] = [L : K]$.
- $\boxed{L|K \text{ is totally ramified}}$ In particular $\lambda = \kappa$ and L is linearly disjoint from K_{nr} (the intersection $L \cap K_{\mathrm{nr}}$ is both totally ramified and unramified, hence it must be K itself). Therefore we have $L_{\mathrm{nr}} = LK_{\mathrm{nr}}$ and $\mathrm{Gal}(L_{\mathrm{nr}}|L) \simeq \mathrm{Gal}(K_{\mathrm{nr}}|K)$. Consider once more the commutative diagram

  diagram

  Indeed, since the extension is totally ramified (is this why?) $v_L(K_{\mathrm{nr}}^\times) = [L : R]v_K(K_{\mathrm{nr}}^\times)$ and restricting to the group itself induces the identity map $\mathrm{Res} = \mathrm{id}$.

□

Theorem 3.2.4 ($\mathrm{Br}(K_{\mathrm{nr}}|K) \simeq \mathrm{Br}(K)$)

Let K be a complete discretely valuated field with finite residue field. We have $\mathrm{Br}(K_{\mathrm{nr}}|K) \simeq \mathrm{Br}(K)$.

Proof: Let $K_n|K$ be the unique unramified extension of K of degree n (this is indeed unique: from Galois correspondence unramified extensions of K correspond $1:1$ to extensions of κ, and for κ finite there is a unique degree n extension). Let $L|K$ be any finite Galois extension of degree n, we shall prove $\mathrm{Br}(L|K) = \mathrm{Br}(K_n|K)$ as subgroups of $\mathrm{Br}(K)$. This concludes as $\mathrm{Br}(K_{\mathrm{nr}}|K)$ is the union of all the $\mathrm{Br}(K_n|K)$ and $\mathrm{Br}(K)$ is the union of all the $\mathrm{Br}(L|K)$ (is this precise? write it in a better way).

- $\boxed{\mathrm{Br}(K_n|K) \subset \mathrm{Br}(L|K)}$ Consider the commutative diagram

  diagram

  From Proposition 3.2.11 the restriction map is in fact multiplication by $[L:K] = n$. Since $\mathrm{Gal}(K_n|K) \simeq \mathbb{Z}/n\mathbb{Z}$ from Corollary 2.2.4 we deduce $[L:K] \cdot \mathrm{Br}(K_n|K) = 0$. Its image in $\mathrm{Br}(L_{\mathrm{nr}}|L) \subset \mathrm{Br}(L)$ is therefore trivial and this shows it is contained in the kernel $\mathrm{Br}(L|K)$.

We now resort to showing $\mathrm{Br}(L|K)$ is finite and has order dividing n. Since it contains $\mathrm{Br}(K_n|K) \simeq \mathbb{Z}/n\mathbb{Z}$ this shows their equality (why? Could it not be $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ for example?). We prove it by steps:

- $\boxed{L|K \text{ is cyclic}}$ From Corollary 3.2.6 we get $\mathrm{Br}(L|K) \simeq \mathbb{Z}/n\mathbb{Z}$.
- $\boxed{\mathrm{Gal}(L|K) \text{ is a } p\text{-group}}$ Let $n = p^r$. Since p-groups are solvable there is a chain of subfields

  $$K = L_0 \subset L_1 \subset \ldots \subset L_r = L$$

  such that $L_i|K$ is Galois and $\mathrm{Gal}(L_i|L_{i-1}) \simeq \mathbb{Z}/n\mathbb{Z}$ for each i. The inflation-restriction sequence (cfr. Proposition 2.2.7) yields

  $$0 \longrightarrow \mathrm{Br}(L_i|L_{i-1}) \xrightarrow{\mathrm{Inf}} \mathrm{Br}(L_i|K) \xrightarrow{\mathrm{Res}} \mathrm{Br}(L_{i-1}|K).$$

  The group $\mathrm{Br}(L_i|L_{i-1})$ has order p and by induction $\mathrm{Br}(L_{i-1}|K)$ has order dividing p^{i-1}, hence $\mathrm{Br}(L_i|K)$ has order dividing p^i. We get $|\mathrm{Br}(L|K)| \mid p^r$.
- $\boxed{\text{General case}}$ Let $n = p_1^{r_1} \ldots p_m^{r_m}$. The Brauer group decomposes as

  $$\mathrm{Br}(L|K) = \bigoplus_{i=1}^{m} \mathrm{Br}(L|K)[p_i].$$

  For each i fix a p_i-Sylow P_i inside $\mathrm{Gal}(L|K)$. Apply Res and Cor to $\mathrm{Br}(L|K)[p_i]$ with respect to the subgroup P_i, we get

  $$\mathrm{Br}(L|K)[p_i] = H^2(\mathrm{Gal}(L|K), L^\times)[p_i] \xrightarrow{\mathrm{Res}} H^2(P_i, L^\times) \xrightarrow{\mathrm{Cor}} H^2(\mathrm{Gal}(L|K), L^\times).$$

  By Proposition 2.2.8 we have $\mathrm{Cor} \circ \mathrm{Res} = [G : P_i]$ and since this is coprime with p_i the composite map is injective. In particular Res itself is injective. We obtain

  $$\mathrm{Br}(L|K) = \bigoplus_{i=1}^{m} \mathrm{Br}(L|K)[p_i] \hookrightarrow \bigoplus_{i=1}^{m} H^2(P_i, L^\times) = \bigoplus_{i=1}^{m} \mathrm{Br}(L|K_i),$$

  where $K_i = L^{P_i}$. By the previous case each component $\mathrm{Br}(L|K_i)$ has order dividing $p_i^{r_i}$ and therefore $|\mathrm{Br}(L|K)| \mid n$.

□

Theorem 3.2.5 (Hasse)

Let K be a complete discretely valuated field with finite residue field κ. There is a canonical isomorphism $\mathrm{inv} : \mathrm{Br}(K) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}$ called the *hasse invariant*. Moreover, if $L|K$ is a finite separable extension the following diagrams commute:

diagrams

Proof: The isomorphism inv is obtained by Theorem 3.2.3 and Theorem 3.2.4. The first commutative diagram is Proposition 3.2.11. It remains to show the second diagram

commutes. Since the multiplication map $[L:K]:\mathbb{Q}/\mathbb{Z}\to\mathbb{Q}/\mathbb{Z}$ is surjective, the restriction map $\mathrm{Res}:\mathrm{Br}(K)\to\mathrm{Br}(L)$ is surjective. Let $\alpha\in\mathrm{Br}(L)$, there is an element $\beta\in\mathrm{Br}(K)$ such that $\alpha=\mathrm{Res}(\beta)$. Moreover, we have $\mathrm{Cor}(\alpha)=\mathrm{Cor}\circ\mathrm{Res}(\beta)=[L:K]\beta$. We get

$$\mathrm{inv}(\alpha)=\mathrm{inv}(\mathrm{Res}(\beta))\overset{(*)}{=}[L:K]\ \mathrm{inv}(\beta)=\mathrm{inv}(\mathrm{Cor}(\alpha)),$$

where $(*)$ follows from the first diagram. □

Corollary 3.2.8

If $(n,\mathrm{char}\ K)=1$ then $H^2(K,\mu_n)=\mathbb{Z}/n\mathbb{Z}$.

Proof: From the exact sequence

$$1\longrightarrow\mu_n\longrightarrow K_s\xrightarrow{n}K_s\longrightarrow 1$$

we extract the long exact sequence in cohomology

$$0=H^1(K,K_s^\times)\longrightarrow H^2(K,\mu_n)\longrightarrow H^2(K,K_s)\xrightarrow{n}H^2(K,K_s).$$

From Theorem 3.2.5 the Brauer group is $H^2(K,K_s)\simeq\mathbb{Q}/\mathbb{Z}$, hence

$$H^2(K,\mu_n)\simeq\ker\left(\mathbb{Q}/\mathbb{Z}\xrightarrow{n}\mathbb{Q}/\mathbb{Z}\right)\simeq\mathbb{Z}/n\mathbb{Z}.$$

□

3.3. The generalised Hilbert symbol

Definition 3.3.25 (Hilbert symbol)

Let $(n,\mathrm{char}\ K)=1$. The *Hilbert symbol* is defined as the map

$$\begin{aligned}H^1(K,\mathbb{Z}/n)\times H^1(K,\mu_n)&\longrightarrow H^2(K,\mu_n\otimes\mathbb{Z}/n),\\ \chi,\qquad\qquad\beta&\longmapsto\chi\smile\beta\end{aligned}$$

Remark

From Kummer theory we can identify $H^1(K,\mu_n)\simeq K^\times/K^{\times n}$. Moreover, from Corollary 3.2.8 we can identify $H^2(K,\mu_n\otimes\mathbb{Z}/n)=H^2(K,\mu_n)\simeq\mathbb{Z}/n$. In this sense the generalised Hilbert symbol can be interpreted as a map

$$H^1(K,\mathbb{Z}/n)\times K^\times/K^{\times n}\longrightarrow\mathbb{Z}/n,$$

that pairs characters of the Galois group and elements of K.

Moreover, when the roots of unity are contained in the field $\mu_n\subset K$ we can fix an isomorphism $\mu_n\simeq\mathbb{Z}/n$ by choosing a root $\omega\in\mu_n$. This gives the generalised Hilbert symbol an even closer resemblance to its original counterpart, as it becomes a map

$$K^\times/K^{\times n}\times K^\times/K^{\times n}\longrightarrow\mu_n.$$

This is effectively a map that pairs two elements of K with values in μ_n. Notice that by fixing the same isomorphism where we substituted $\mu_n\simeq\mathbb{Z}/n$ the final map does not depend on the choice of ω.

Proposition 3.3.12

Suppose $\chi \in H^1(K, \mathbb{Z}/n\mathbb{Z})$ corresponds to a surjective character $\mathrm{Gal}(K_s|K) \twoheadrightarrow \mathbb{Z}/n\mathbb{Z}$ (by eventually making n smaller we can always find such a surjective character), let L be the fixed field for $\ker \chi$. If $b \in K^\times$ then

$$(\chi, b) = 1 \iff b \in N_{L|K}(L^\times).$$

From this proposition we recover the classical Hilbert symbol, making the generalised Hilbert symbol effectively a generalisation of the former. If $n = 2$ we have $\{\pm 1\} = \mu_n \subset K$ and $L = K(\sqrt{a})$ for some $a \in K$. We get

$$\begin{aligned}(\chi, b) = 1 &\iff b \in N_{L|K}(L^\times) \\ &\iff b = N\left(x + \sqrt{a}y\right) = x^2 + ay^2 \\ &\iff x^2 + ay^2 - bz^2 = 0 \text{ has a non-trivial solution}\end{aligned}$$

that is exactly the definition of the classical Hilbert symbol when $K = \mathbb{Q}_p$.

Proof:

□